

Functions of OS, process management and GUI ★ Golden

One college period · 40–45 minutes · Operating System

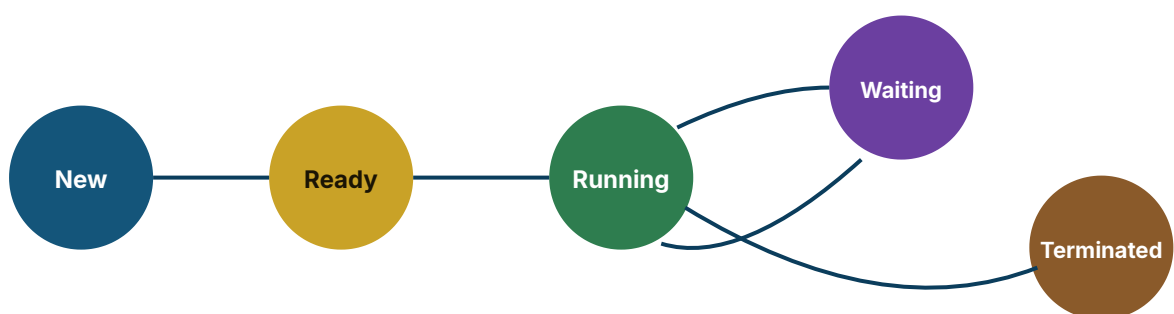
What you will learn

- List and explain the functions of an OS
- Define process and thread; draw process states
- Use files, folders, Control Panel, msconfig, dxdiag at a theory level
- Mention open-source OS and mobile OS

Functions of an operating system

Function	What the OS does
Booting	POST, load kernel, start system services
Process management	Create, schedule, suspend, terminate processes
Memory management	Allocate RAM, virtual memory / paging
File management	Create, copy, delete, rename files and folders; FAT/NTFS
I/O management	Talk to keyboard, disk, printer via drivers
Secondary storage management	Free space, disk scheduling
Network management	TCP/IP, sharing, firewall basics
Protection / security	Passwords, permissions, user accounts
Command interpreter / shell	cmd, PowerShell, Bash, or GUI Explorer

Process. A program in execution. A thread is a lightweight unit of a process that shares the same memory space. Multi-threading: many threads in one process. Multi-tasking: many processes.



Process states: New → Ready → Running → Waiting or Terminated.

- New: being created.
- Ready: waiting for CPU.
- Running: on the CPU.
- Waiting / blocked: waiting for I/O or an event.
- Terminated: finished or killed.

Working with a GUI OS (Windows)

Create / delete / copy / rename files; search; attributes (Read-only, Hidden, Archive, System); drives and paths (C:\Users\Ali\Notes.txt); Control Panel / Settings for hardware; dxdiag to list display and sound; msconfig to change startup programs.

Open-source OS (Linux, Ubuntu) can be studied, copied and improved freely. Mobile OS: Android versions, iOS.

HOW BIEK ASKS THIS

Long answer: OS definition + at least 7 functions with one line each. Short: process vs thread; five process states.

اردو میں سمجھیں

OS بوٹ کرتا ہے، پروسیس اور میموری بانٹتا ہے، فائلیں سنبھالتا ہے، پرنٹر چلاتا ہے اور پاس ورڈ سے حفاظت کرتا ہے۔
پروسیس چلتا ہوا پروگرام ہے۔ حالتیں: نیا، تیار، چل رہا، انتظار، ختم۔

CHECK YOURSELF

1. Name five OS functions.
2. Ready vs running?

Answers: 1. Booting, process, memory, file, I/O (any five) · 2. Ready waits for CPU; running currently has the CPU

MCQs

1. A program in execution is called a:
 - a. File
 - b. Process
 - c. Folder
 - d. Port
2. File management is a function of the:
 - a. Compiler
 - b. Operating system
 - c. Browser
 - d. ALU

Key: 1=b, 2=b

Short questions (3 marks each)

1. Explain the functions of an operating system.
2. Differentiate process and thread.
3. Write the states of a process.

Model short answers

1. Booting, process, memory, file, I/O, storage, network, protection, command interpreter — each in one sentence as in the table.
2. Process: whole program in execution with its own memory. Thread: a path inside a process sharing memory; lighter to create.
3. New, Ready, Running, Waiting/Blocked, Terminated.

Long questions (board Section C)

1. What is an Operating System? Explain its functions. Also describe process management.